

SEIF ZAKARIA

Full-stack Software Engineer – Cairo, Egypt

+201558173354 | seifzellaban@gmail.com | linkedin.com/in/seifzellaban | github.com/seifzellaban | seifzellaban.wiki

Experience

Safety Source

Dec. 2025 – Feb. 2026

Freelance Full-Stack Engineer

Cairo, Egypt

- Engineered an admin management system tracking **7M+ EGP** in monthly sales across multiple categories for a 12-person sales team, using **Next.js**, **Convex**, and **Tailwind CSS**.
- Designed a centralized real-time analytics dashboard, replacing fragmented spreadsheet workflows and becoming the single source of truth for all operational data.

Masons

Sept. 2024 – Present

Founder & Lead Software Engineer

Cairo, Egypt

- Founded a **pre-seed AI software studio** building production products for real users, currently shipping **Rafiqi**, an LLM-powered mental health platform with culturally aware UX built specifically for Egyptians.
- Built **AgencyKit**, a SaaS platform for agencies to manage clients, workspaces, and content workflows with WorkOS-powered auth and role-based permissions, using **Next.js**, **TypeScript**, **Convex**, and **Tailwind CSS**.

DEPI x EYouth

Nov. 2024 – May 2025

Full Stack Web Development Training (MERN Stack)

Cairo, Egypt

- Architected the Node.js/Express.js backend for **Convo**, a real-time chat application with WebSocket support (Socket.io), designed to scale to 500+ concurrent users without slowing down.
- Implemented a secure, stateless RESTful API with JWT-based authentication and led the backend team's agile workflow, including code reviews, Git branching strategy, and automated testing.

Projects

Organum | *Python, NumPy, PyQt6, sounddevice*

Feb. 2026

- Built a real-time pipe organ synthesizer from scratch using additive synthesis and physically modeled pipe acoustics: 11 organ stops, 9 Hammond-style drawbars, stereo Schroeder cathedral reverb, and 24-voice polyphony.
- Engineered a fully vectorized NumPy audio pipeline with band-limited wavetables, pre-allocated work buffers eliminating 1,000 allocations per audio callback, and a lock-free event queue within a 93ms real-time budget.

Umbra | *Next.js, D3.js, Gemini API, shadcn/ui, Tailwind*

Oct. 2025

- Enabled semantic exploration of 608 NASA space biology papers via D3.js knowledge graph and Gemini-powered contextual search, connecting 50 years of fragmented research in a single interface.

repAi | *Angular, Gemini API, Firebase, GCP*

Feb. 2025

- Built an AI recycling rewards platform with real-time material detection and dynamic cashback pricing via Gemini API. Placed **Top 5 of 88 teams** at GDG Cairo AI Finance Hackathon 2025.

Orbit | *Next.js, Three.js, NASA APIs*

Oct. 2024

- Built an interactive 3D solar system simulation with real-time asteroid tracking via NASA's Small Body Database, optimizing Three.js shader rendering for 60 FPS performance.

Education

Ain Shams University

Cairo, Egypt

B.Sc. in Computer Science and Statistics

Expected 2027

- Relevant Coursework:** Data Structures, Algorithms Analysis, Design Patterns, Database Management, Networking, Systems Programming, Probability Theory, Statistical Inference, Linear Algebra

Relevant Skills

Frontend: JavaScript, TypeScript, React, Next.js, Angular, Tailwind CSS, D3.js, shadcn/ui | **Backend & DBs:** Node.js, Express.js, Python, PostgreSQL, MySQL, MongoDB, Firebase, Convex, Drizzle ORM | **Tools:** Git, Linux, Socket.io, GCP, RESTful APIs, JWT, WorkOS, NumPy, Agile (Scrum) | **Languages:** Arabic (Native), English (C1)

Leadership & Extracurricular

Microsoft Student Partner

Sept. 2025 – Present

Head of AI & Resonance (35+ Committee Members)

Ain Shams University

- Designed a 7-month audio AI curriculum across 18 structured sessions, covering signal processing, deep learning, and capstone projects including a Shazam replica and piano transcription system.

Open Source Community

Nov. 2023 – Sept. 2025

Head of HR & Public Relations Team Lead (130+ Organization Members)

Ain Shams University

- Streamlined recruitment pipeline through a faster interview protocol, increasing interview throughput by **94%** (54 personally conducted), and authored the community's HR policy guidelines.
- Authored and published content across community platforms, led 15 internal workshops on talent assessment, and delivered 2 keynote speeches to 300+ incoming freshmen.